Convergence Simulations Model B

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1 Model Structure

1.1 Data and indexing

The data are N binary trials. Trial i has

- a response $y_i \in \{0, 1\}$, the monkey's categorization of the stimulus as "low" (0) or "high" (1);
- a standardized stimulus amplitude x_i. The raw amplitudes are centered and scaled by the mean and standard deviation of the distinct amplitude levels. The covariate row is $(1, x_i)$;
- a group $g[i] \in \{1, \ldots, 4\}$: the combination of hand and distractor condition,

  $$g \in \{\text{left_uni},\ \text{left_bi},\ \text{right_uni},\ \text{right_bi}\},$$

  where "uni" is unimanual (no distractor) and "bi" is bimanual (distractor present);
- a session $s[i] \in \{1, \ldots, S\}$; the real data have $S = 42$.

1.2 Likelihood

Each response is Bernoulli,

$$y_i \sim \text{Bernoulli}(\psi_i), \qquad \psi_i = \psi\big(x_i;\ \theta_{g[i],s[i]}\big),$$

with the psychometric function

$$\psi(x; \theta) = \gamma_h + (1 - \gamma_h - \gamma_l)\ \text{logit}^{-1}(\beta_0 + \beta_1 x), \qquad \theta = (\beta_0, \beta_1, \gamma_h, \gamma_l).$$

Here β_0 and β_1 are parameters of the sigmoid, and γ_h, γ_l set the asymptotes:

$$\lim_{x \to -\infty} \psi = \gamma_h, \qquad \lim_{x \to +\infty} \psi = 1 - \gamma_l.$$

So γ_h is the probability of responding 1 ("high") at very low amplitudes, and γ_l the probability of responding 0 ("low") at very high amplitudes. In the code ψ_i is clipped to $[10^{-8}, 1 - 10^{-8}]$ before entering the Bernoulli likelihood, for numerical stability.

1.3 Session-level

For each group g and session s, the four parameters are drawn independently from group-specific distributions:

$$\begin{aligned}
\beta_{0,gs} &\sim \mathcal{N}\big(\mu_{\beta_0,g},\ \sigma_{\beta_0,g}\big),\\
\beta_{1,gs} &\sim \mathcal{N}\big(\mu_{\beta_1,g},\ \sigma_{\beta_1,g}\big),\\
\gamma_{h,gs} &= 0.25\, w_{h,gs}, \qquad w_{h,gs} \sim \text{LogitNormal}\big(\mu_{\gamma_h,g},\ \sigma_{\gamma_h,g}\big),\\
\gamma_{l,gs} &= 0.25\, w_{l,gs}, \qquad w_{l,gs} \sim \text{LogitNormal}\big(\mu_{\gamma_l,g},\ \sigma_{\gamma_l,g}\big).
\end{aligned}$$

The LogitNormal keeps $w \in (0, 1)$, so each lapse parameter γ lies in $(0, 0.25)$. That range keeps the curve increasing and makes the PSE and JND below well defined.

Groups are independent of each other (no pooling across groups).

1.4 Group-level Hyperpriors

Each of the four groups has its own 8 hyperparameters, with the same priors in every group:

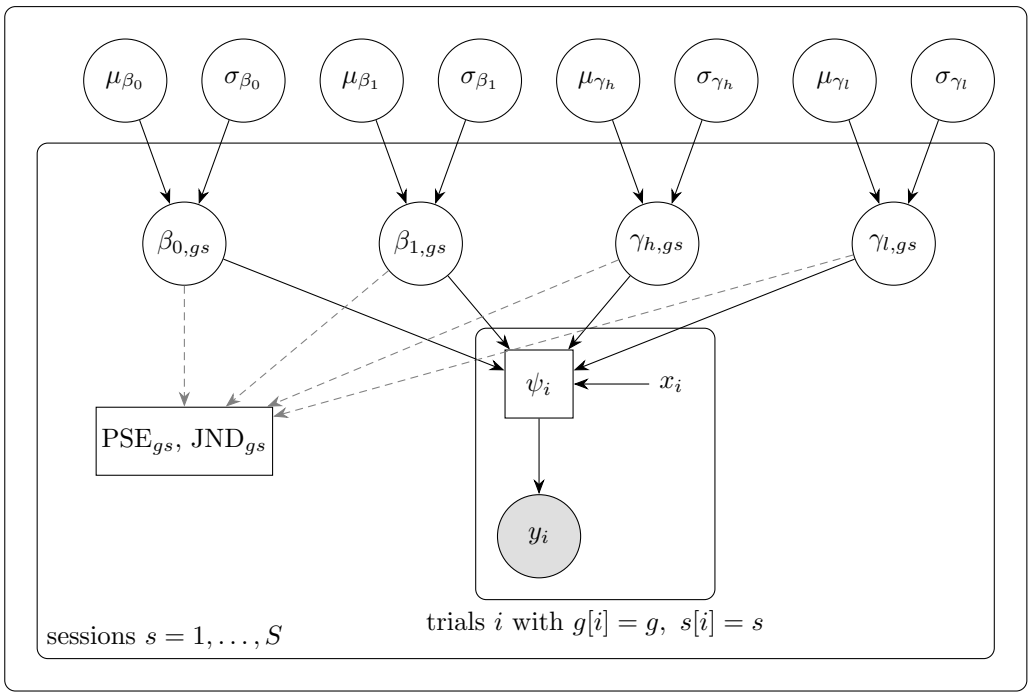
$$\begin{aligned}
\mu_{\beta_0,g} &\sim \mathcal{N}(0,\ 2), & \sigma_{\beta_0,g} &\sim \text{HalfNormal}(2),\\
\mu_{\beta_1,g} &\sim \text{LogNormal}(\log 4,\ 0.5), & \sigma_{\beta_1,g} &\sim \text{Exponential}(\lambda = 1) \quad \text{(mean 1)},\\
\mu_{\gamma_h,g},\ \mu_{\gamma_l,g} &\sim \mathcal{N}(-3,\ 1.5), & \sigma_{\gamma_h,g},\ \sigma_{\gamma_l,g} &\sim \text{HalfNormal}(0.5).
\end{aligned}$$

All hyperparameters are independent. The priors were chosen from the stimulus design and general knowledge of the task, and checked by prior predictive simulation (`Check_Prior_B.py`), not from fits to the real data:

- μ_{β_0}: the stimulus amplitudes (6 to 50) are symmetric around 28, the category boundary, which standardizes to $x = 0$. This prior puts a group's typical PSE between amplitudes 13 and 43 with 90% probability, and outside the stimulus range only 3% of the time.
- σ_{β_0}: weakly informative, bounded by the design. The implied day-to-day sd of the PSE has median about 5 amplitude units and is below about 20 with 95% probability, small next to the stimulus range.
- μ_{β_1}: positive with no upper bound; median 4, 95% in [1.5, 10.6].
- σ_{β_1}: mean 1, so that sessions vary around the group slope without the session slopes turning negative (2% prior probability).
- μ_γ (logit scale): the typical lapse rate $0.25\ \text{logit}^{-1}(\mu_\gamma)$ has median 0.012 and 95% range [0.0007, 0.12]; trained animals lapse on a few percent of trials and rarely more than about 10%.
- σ_γ (logit scale): median 0.34. For a group whose typical lapse rate is 0.05, this puts about 80% of sessions between 0.03 and 0.07.

An earlier version of the model used $\mu_{\beta_0} \sim \mathcal{U}[-12,\ 12]$, $\sigma_{\beta_0} \sim$ Exponential(mean 12), $\mu_{\beta_1} \sim \mathcal{U}[0,\ 8]$, $\sigma_{\beta_1} \sim$ Exponential(mean 4), $\mu_\gamma \sim \mathcal{N}(-2.5,\ 1)$ and $\sigma_\gamma \sim$ HalfNormal(1.25). Its convergence study motivated the changes (Section 3.6).

1.5 Graphical Model



Circles are random variables (shaded: observed), rectangles are deterministic functions, and x_i is a fixed covariate. Dashed arrows lead to derived quantities that do not feed back into the likelihood.

1.6 Derived Quantities: PSE and JND

For each group and session the model records two summaries of the psychometric curve.

Point of subjective equality (PSE): the stimulus at which a response of "high" and "low" are equally likely ($\psi = 1/2$). Solving $\psi(x;\theta) = \frac{1}{2}$ for x gives

$$\text{PSE} = \frac{1}{\beta_1}\left[-\beta_0 + \log\frac{1-2\gamma_h}{1-2\gamma_l}\right].$$

Just noticeable difference (JND): the amount the stimulus must be changed for the difference to be noticeable half of the time, a measure of sensitivity. This is computed as half the stimulus distance between the points where $\psi = 0.25$ and $\psi = 0.75$,

$$\text{JND} = \frac{x_{.75} - x_{.25}}{2} = \frac{1}{2\beta_1}\,\log\frac{(3-4\gamma_h)(3-4\gamma_l)}{(1-4\gamma_h)(1-4\gamma_l)}.$$

Both are in units of the standardized stimulus. When $\gamma_h = \gamma_l = 0$, they reduce to $\text{PSE} = -\frac{\beta_0}{\beta_1}$ and $\text{JND} = \frac{\log 3}{\beta_1}$.

1.7 Implementation

To aid fit, as in the convention of typical logistic regression, the raw amplitudes x_i are centered and scaled by the mean and standard deviation of the distinct amplitude levels. To aid convergence of HMC NUTS, all distributions are written as deterministic transforms of a standard normal or uniform random variable. These non-centered parameterizations mean the sampler only works with uniform or standard normal variables.

2 Synthetic Data Generation

The true hyperparameters are the posterior means of Model B's fit to the real data (`Results_B.pkl`, from `Main_Analysis_B.py`; see the report "Model B Fit to the Real Data"), so the synthetic data resemble the real data (Table 1). They only generate data; the priors do not use them. Each replicate draws new session-level parameters from them.

Table 1: True hyperparameters: posterior means of the real-data fit. The lapse-rate ones are on the logit scale; the typical lapse rate is 0.25 $\text{logit}^{-1}(\mu_\gamma)$.

	left_uni	left_bi	right_uni	right_bi
μ_{β_0}	−1.487	−0.099	−1.056	0.686
μ_{β_1}	4.968	4.321	4.580	4.340
μ_{γ_h}	−5.203	−2.350	−4.729	−1.816
μ_{γ_l}	−3.445	−3.442	−3.717	−3.228
σ_{β_0}	0.763	0.772	1.011	0.759
σ_{β_1}	0.285	1.266	0.173	0.615
σ_{γ_h}	0.359	0.591	0.371	0.362
σ_{γ_l}	0.373	0.393	0.381	0.431

Compared with the earlier, hand-picked values (Section 3.6), these have group slopes of 4.3–5.0 rather than 3 and 6, between-session sds of the slope that are far from 0 (up to 1.27 for left_bi), and unimanual γ_h close to 0 (typical values 0.001–0.002).

The model was fit to 10 dataset conditions in two sweeps, each run twice with new data:

- the sessions sweep varies the number of sessions (5, 10, 20, 40 and 80, each with 308 trials), mainly testing the hyperparameters;
- the trials sweep varies the trials per session (20, 40, 80, 160 and 320, with 42 sessions), mainly testing the session-level parameters.

Within a replicate the datasets of a sweep are nested (the first n sessions, or the first t trials of every session). The real data have 42 sessions with about 308 trials each, so the 40-session and 320-trial fits are the most representative.

3 Model Fit Results

Every figure and number in this report is produced by `Convergence_B.py`; the cell "Numbers for Convergence_B_Report.tex" prints the numbers that the earlier cells do not, and also runs on the saved results without refitting. The statements about what the priors imply come from

`Check_Prior_B.py`. Hyperparameter entries are averaged over the two replicates; session-level RMSEs and coverages are pooled over them (84 sessions per group in the trials sweep). The fits sample the 4 chains in parallel; the whole study takes 0.7 hours.

3.1 Sampling Diagnostics

All 20 fits sampled without divergences, with $\hat{R} \leq 1.011$ and a bulk ESS of at least 670 for every hyperparameter, and $\hat{R} \leq 1.007$ for every session-level parameter.

Table 2: Sampler diagnostics over both replicates. $\hat{R}$ and bulk ESS are taken over the hyperparameters.

Sweep	Size	max $\hat{R}$	min ESS	divergences
sessions	5	1.003	1503	0
sessions	10	1.005	1297	0
sessions	20	1.006	909	0
sessions	40	1.007	758	0
sessions	80	1.011	801	0
trials	20	1.004	961	0
trials	40	1.005	1074	0
trials	80	1.006	822	0
trials	160	1.007	1000	0
trials	320	1.004	670	0

3.2 Hyperparameter means

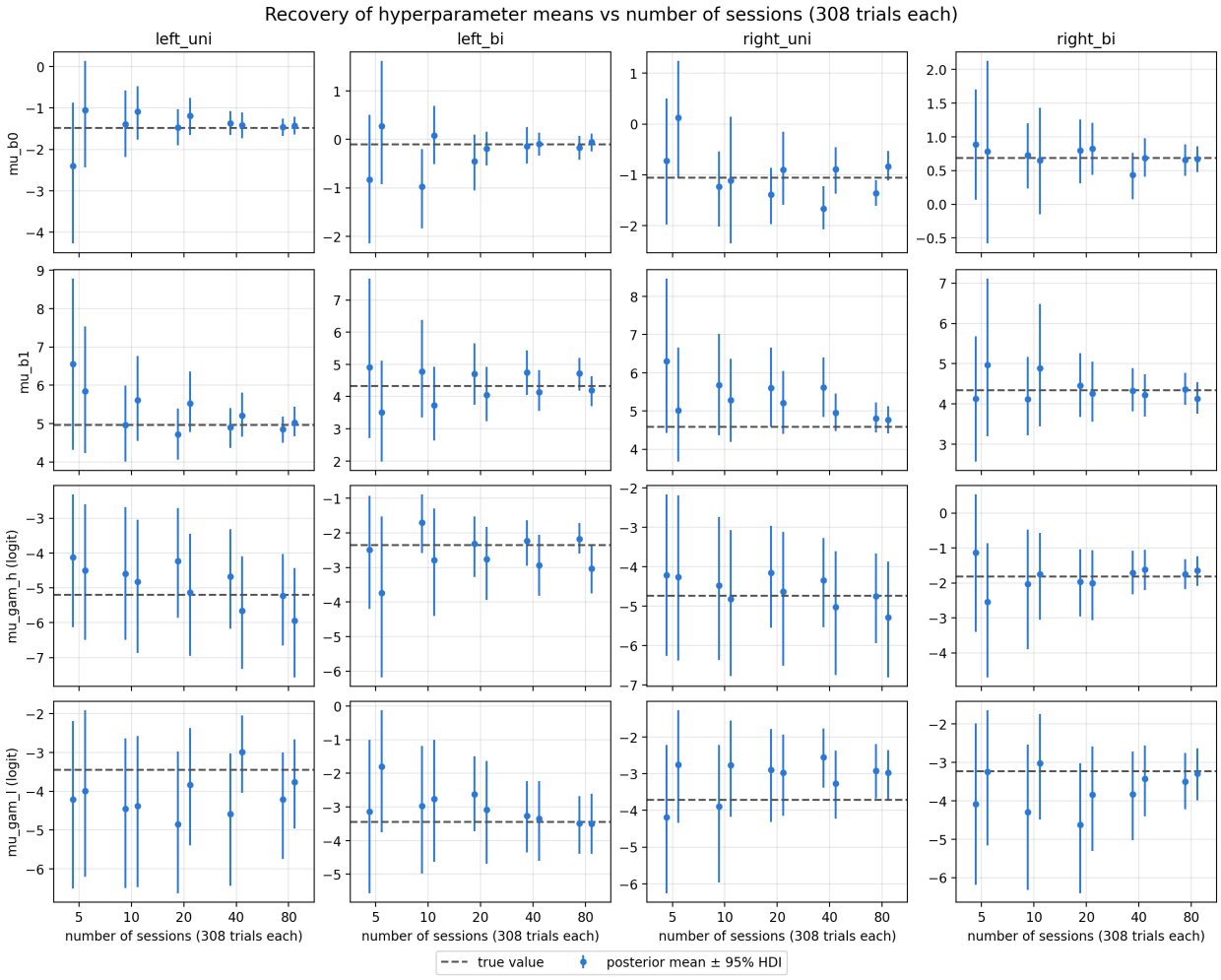


Figure 1: Posterior mean and 95% HDI of the group means against the number of sessions (308 trials each). The two replicates are drawn side by side; the dashed line is the true value.

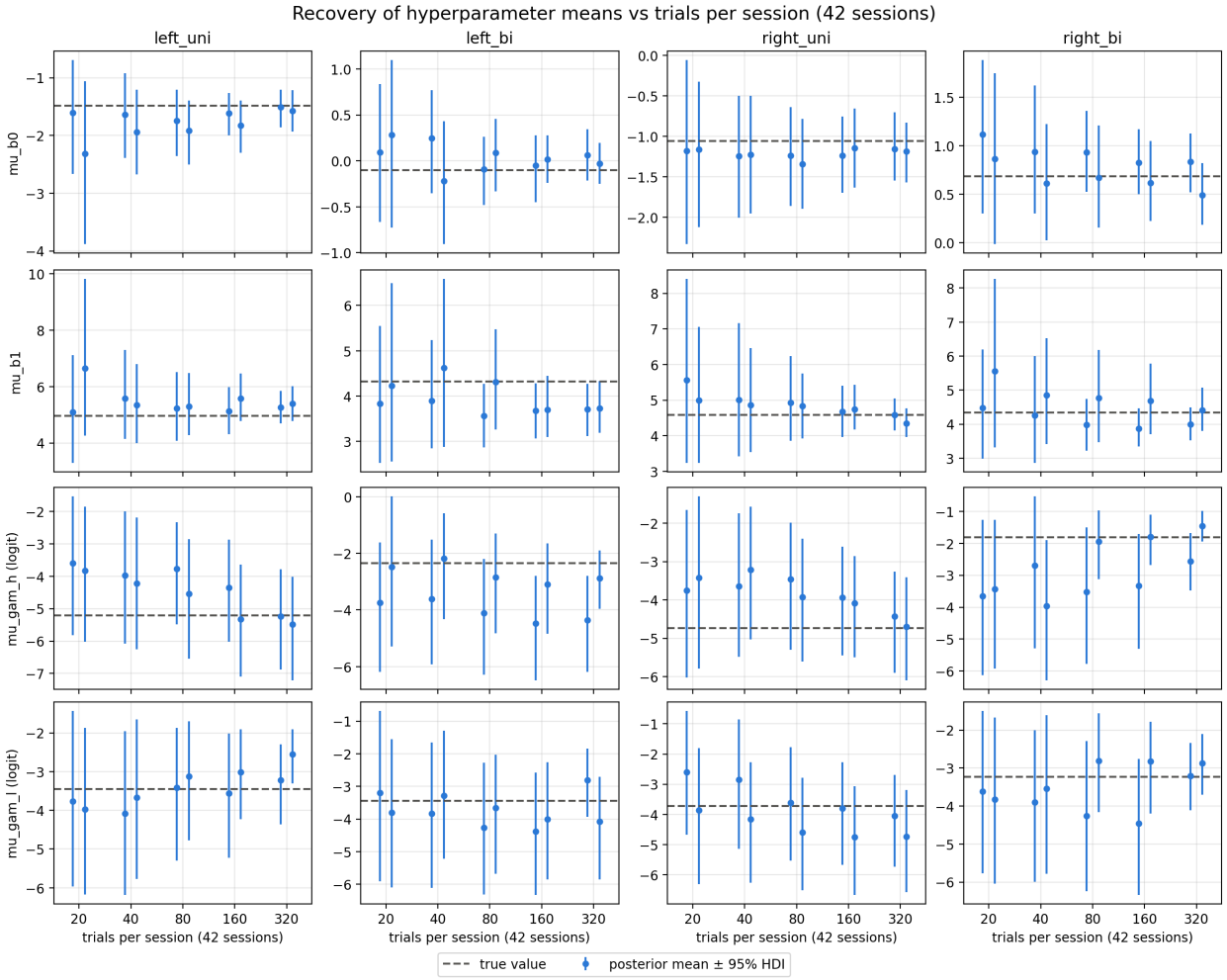


Figure 2: As Figure 1, against trials per session (42 sessions).

Table 3: Group means at 40 and 80 sessions (sessions sweep): true value and posterior mean [95% HDI], averaged over replicates. Lapse-rate means are on the logit scale.

		left_uni	left_bi	right_uni	right_bi
μ_{β_0}	true	−1.49	−0.10	−1.06	0.69
	40	−1.40 [−1.70, −1.09]	−0.12 [−0.42, 0.20]	−1.28 [−1.72, −0.84]	0.56 [0.24, 0.87]
	80	−1.44 [−1.66, −1.23]	−0.11 [−0.34, 0.10]	−1.10 [−1.36, −0.82]	0.66 [0.45, 0.87]
μ_{β_1}	true	4.97	4.32	4.58	4.34
	40	5.06 [4.51, 5.61]	4.44 [3.81, 5.13]	5.29 [4.66, 5.93]	4.27 [3.76, 4.82]
	80	4.94 [4.58, 5.31]	4.46 [3.95, 4.92]	4.79 [4.42, 5.18]	4.25 [3.87, 4.65]
μ_{γ_h}	true	−5.20	−2.35	−4.73	−1.82
	40	−5.17 [−6.75, −3.70]	−2.58 [−3.38, −1.84]	−4.68 [−6.13, −3.43]	−1.66 [−2.26, −1.07]
	80	−5.58 [−7.11, −4.23]	−2.60 [−3.18, −2.02]	−5.01 [−6.37, −3.76]	−1.70 [−2.13, −1.28]
μ_{γ_l}	true	−3.44	−3.44	−3.72	−3.23
	40	−3.79 [−5.24, −2.54]	−3.31 [−4.48, −2.24]	−2.91 [−3.80, −2.08]	−3.63 [−4.71, −2.64]
	80	−3.99 [−5.35, −2.83]	−3.49 [−4.40, −2.64]	−2.95 [−3.68, −2.28]	−3.40 [−4.11, −2.70]

Over all 20 fits, the HDIs of the group means contain the true value in 94–96% of cases (Table 3).

Bias μ_{β_0}: Recovered in all groups. At 40 sessions the posterior means are within 0.22 of the truth, with HDI widths of 0.61–0.88. For right_uni one of the two replicates misses the truth at 40 and at 80 sessions.

Slope μ_{β_1}: Recovered in all groups. At 80 sessions the posterior means are within 0.21 of the truth and every HDI contains it. At 40 sessions the HDIs are 1.06–1.32 wide, and right_uni lies above the truth (5.29 against 4.58, one replicate missing it). Unlike the earlier study, where a true slope of 6 was poorly identified, the real slopes of 4.3–5.0 make the rise from $\psi = 0.25$ to 0.75 span more than one stimulus spacing.

Lapse-rate means μ_{γ_h} and μ_{γ_l}: The least precise means, because lapses are rare and show only where the sigmoid has reached its floor or ceiling (about 3 expected γ_l lapses per stimulus level for the left-hand groups at the real data size). Their HDIs contain the truth in nearly every fit, but three points stand out:

- μ_{γ_h} of the unimanual groups, whose true typical γ_h is 0.001–0.002, has wide HDIs (2.7–3.1 at 40 sessions): with almost no lapses the data only say that γ_h is small.
- μ_{γ_h} of left_bi is too low in the trials sweep: -3.62 [-5.07, -2.35] at 320 trials against a true -2.35. Its curve never reaches its floor within the stimulus range.
- μ_{γ_l} of right_uni is too high: at 80 sessions -2.95 [-3.68, -2.28] against a true -3.72, and both replicates' HDIs miss the truth. The earlier study found the same overestimate for this group, under both sets of priors and at different true values; its cause has not been identified.

μ_{γ_l} of left_uni is also too high in the trials sweep (-2.88 against -3.44 at 320 trials), though its HDI contains the truth.

3.3 Hyperparameter standard deviations

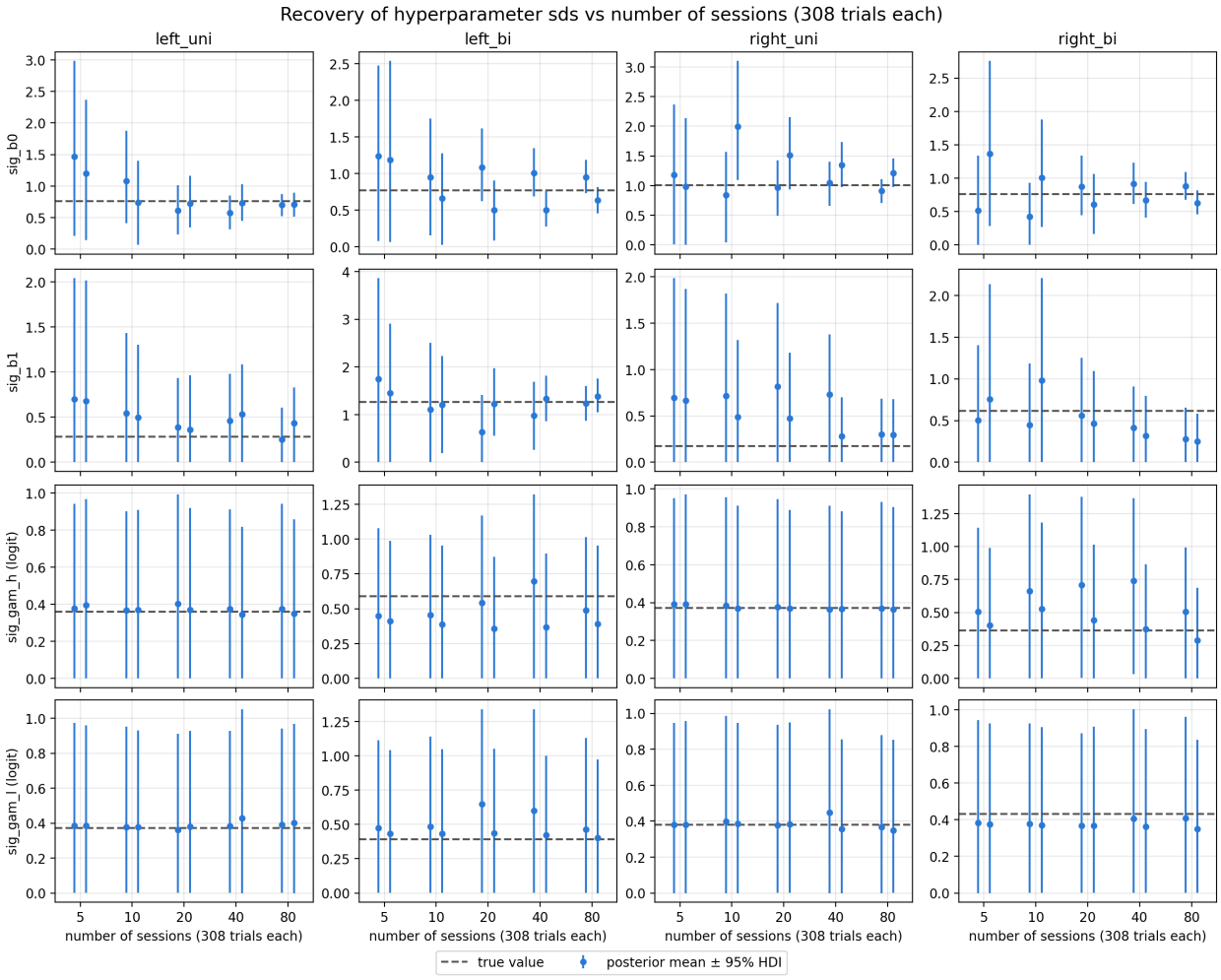


Figure 3: Posterior mean and 95% HDI of the between-session standard deviations against the number of sessions.

Table 4: Between-session sds at 80 sessions: true value and posterior mean [95% HDI], averaged over replicates.

	left_uni	left_bi	right_uni	right_bi
σ_{β_0} true	0.76	0.77	1.01	0.76
posterior	0.70 [0.52, 0.88]	0.79 [0.59, 1.00]	1.06 [0.84, 1.29]	0.75 [0.56, 0.95]
σ_{β_1} true	0.29	1.27	0.17	0.62
posterior	0.34 [0.00, 0.72]	1.31 [0.96, 1.68]	0.30 [0.00, 0.68]	0.26 [0.00, 0.62]
σ_{γ_h} true	0.36	0.59	0.37	0.36
posterior	0.36 [0.00, 0.90]	0.44 [0.00, 0.99]	0.37 [0.00, 0.92]	0.40 [0.00, 0.84]
σ_{γ_l} true	0.37	0.39	0.38	0.43
posterior	0.40 [0.00, 0.96]	0.43 [0.00, 1.05]	0.36 [0.00, 0.86]	0.38 [0.00, 0.90]

σ_{β_0}: Recovered at 80 sessions in every group (Table 4). With fewer sessions it can be well off: for right_uni the posterior mean is 1.42, 1.24 and 1.19 at 10, 20 and 40 sessions against a true 1.01.

σ_{β_1}: Recovered when it is large: for left_bi (true 1.27) the posterior is 1.31 [0.96, 1.68] at 80 sessions. The smaller values (0.17–0.62) cannot be told apart from 0 at these data sizes: their HDIs start at 0, and for right_bi the posterior mean falls to 0.26 at 80 sessions against a true 0.62, with the HDI just reaching the truth.

σ_{γ_h}, σ_{γ_l}: Not identified. Their posterior means are 0.35–0.59 in every fit, their HDIs start at 0 and end at 0.84–1.29, and they barely change with the amount of data: they reflect the prior (mean 0.40, 95% point 0.98). The true values (0.36–0.59) happen to lie near the prior median (0.34), so the 100% coverage says little.

3.4 Session-level parameters

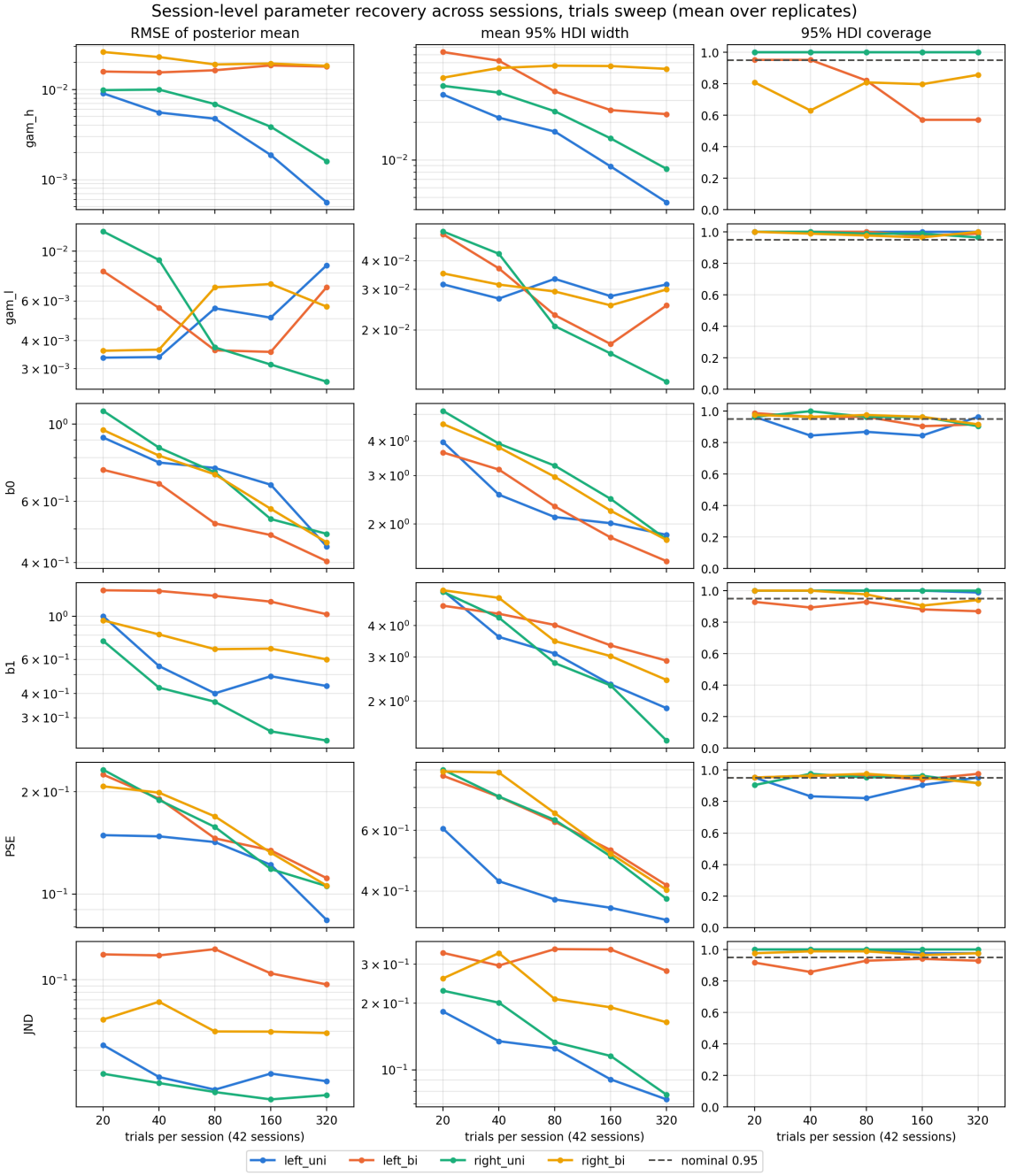


Figure 4: Session-level recovery against trials per session (42 sessions), averaged over replicates. Left: RMSE of the posterior mean across sessions. Middle: mean 95% HDI width. Right: fraction of sessions whose HDI contains the true value.

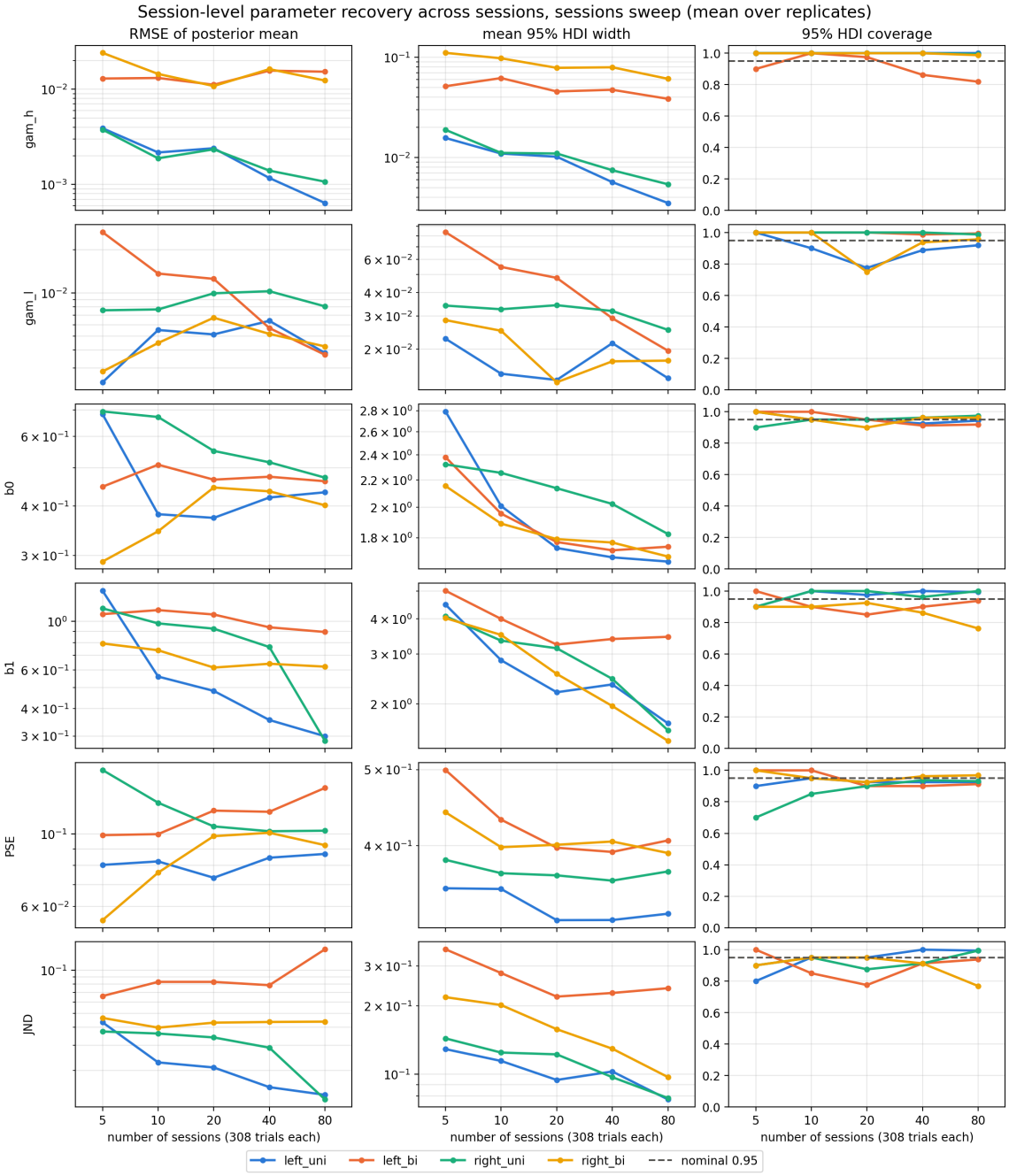


Figure 5: As Figure 4, against the number of sessions (308 trials each).

Table 5: Session-level recovery at 320 trials per session (42 sessions, both replicates): RMSE of the posterior mean / mean HDI width / coverage. PSE and JND in standardized units (1 unit = 14.63 μm).

	left_uni	left_bi	right_uni	right_bi
PSE	0.08 / 0.33 / 0.95	0.11 / 0.42 / 0.98	0.11 / 0.38 / 0.92	0.11 / 0.40 / 0.92
JND	0.017 / 0.07 / 0.98	0.095 / 0.28 / 0.93	0.013 / 0.08 / 1.00	0.039 / 0.16 / 0.98
β_0	0.45 / 1.83 / 0.96	0.40 / 1.47 / 0.92	0.49 / 1.76 / 0.90	0.46 / 1.75 / 0.92
β_1	0.44 / 1.88 / 0.99	1.02 / 2.90 / 0.87	0.23 / 1.39 / 1.00	0.60 / 2.43 / 0.94
γ_h	0.001 / 0.005 / 1.00	0.018 / 0.023 / 0.57	0.002 / 0.008 / 1.00	0.018 / 0.054 / 0.86
γ_l	0.009 / 0.031 / 1.00	0.008 / 0.026 / 0.99	0.003 / 0.012 / 0.96	0.006 / 0.030 / 1.00

PSE and JND. The per-session PSEs are recovered well: at 320 trials the RMSE is 0.08–0.11 standardized units (1.2–1.6 μm) and the coverage 0.92–0.98 (Table 5), down from an RMSE of 0.15–0.23 at 20 trials. The JND is recovered with an RMSE of 0.013–0.039 in three groups. For left_bi, whose sessions differ strongly in slope ($\sigma_{\beta_1} = 1.27$, true session JNDs from 2.4 to 15.9 μm), the RMSE is 0.095 (1.4 μm), with coverage 0.93.

β_0, β_1. The individual coefficients are less precise than the PSE and JND computed from them (HDI widths 1.4–2.9 at 320 trials). Coverage is 0.90–1.00, except for the left_bi slopes (0.87), the group with the largest slope spread.

Coverage. With 84 sessions per group, the standard error of a coverage estimate near 0.95 is about 0.024, so values of 0.90–0.92 are mildly low.

Lapse rates γ_h and γ_l: a miss at the group level carries over to every session. With σ_{γ_h} and σ_{γ_l} not identified, the session lapse rates are pulled strongly toward their group's typical value. The session HDIs are therefore narrow (mean width 0.005–0.054), and when the group value is off, every session of the group misses:

- γ_h of the bimanual groups is undercovered at 320 trials: 0.57 for left_bi and 0.86 for right_bi. In one replicate the left_bi session posterior means average 0.005 against true values averaging 0.021 (coverage 0.31), and the right_bi ones 0.021 against 0.037 (coverage 0.71). In the other replicate right_bi goes the other way (0.049 against 0.036).
- γ_l of left_uni is too high at 320 trials (posterior means averaging 0.016 against 0.0085), the counterpart of the μ_{γ_l} error above; its HDIs still contain the truth.

3.5 Posterior densities

The following plots show the same fits as full posterior densities, darker for larger datasets. They confirm the points above: the group means and σ_{β_0} concentrate around the truth as data grow, while the posteriors of σ_{γ_h} and σ_{γ_l} barely change.

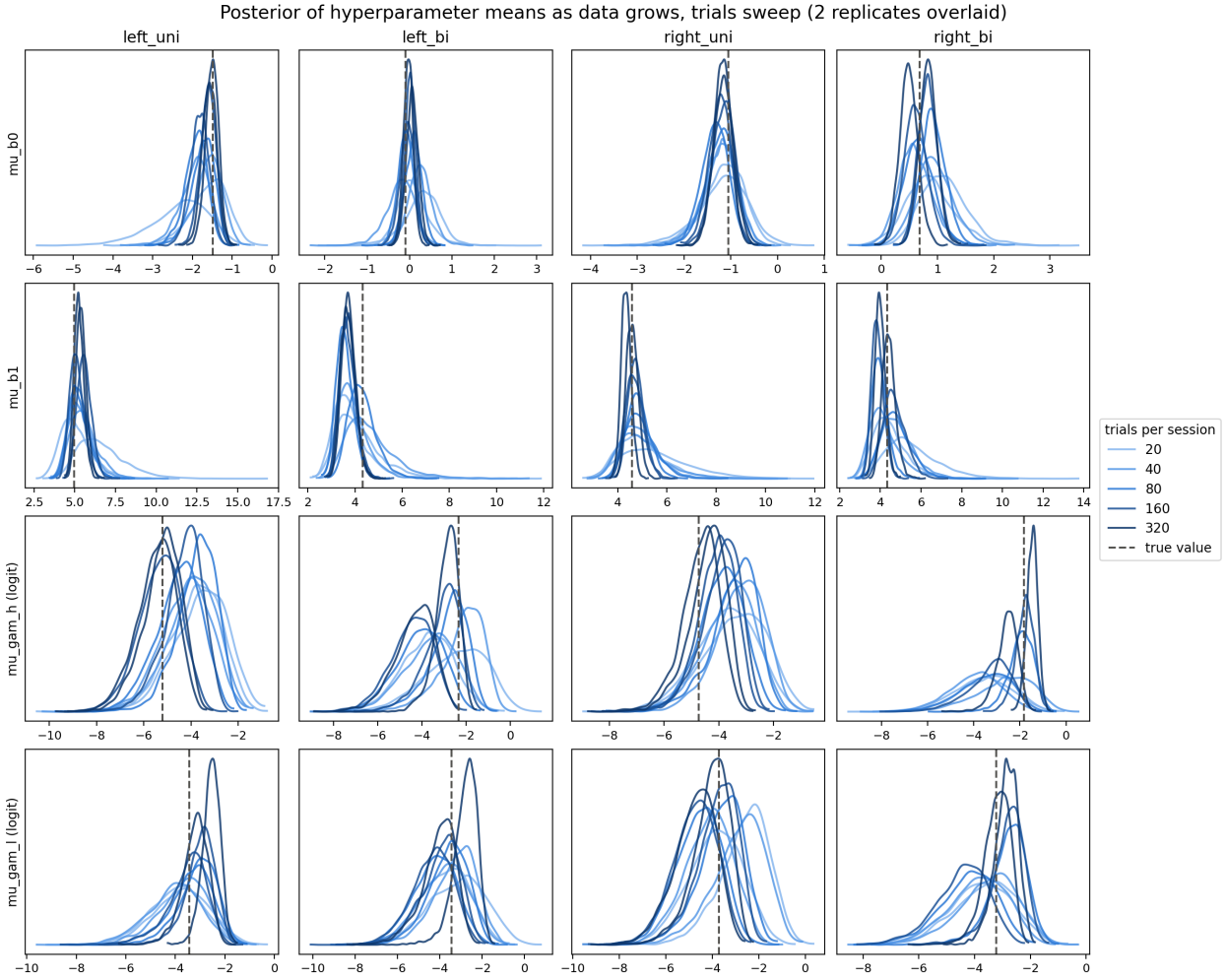


Figure 6: Posteriors of mean hyperparameters as trials vary

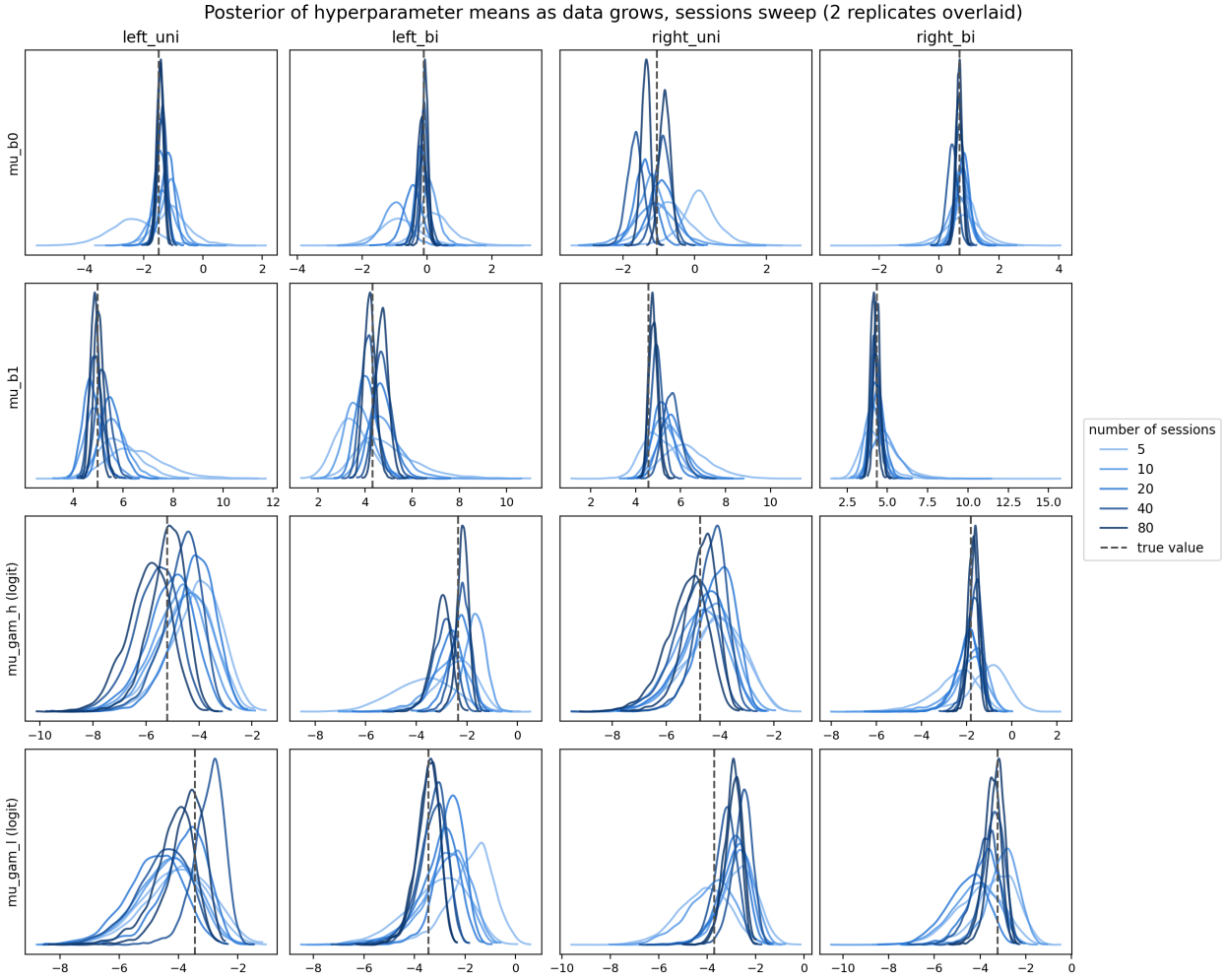


Figure 7: Posteriors of mean hyperparameters as sessions vary

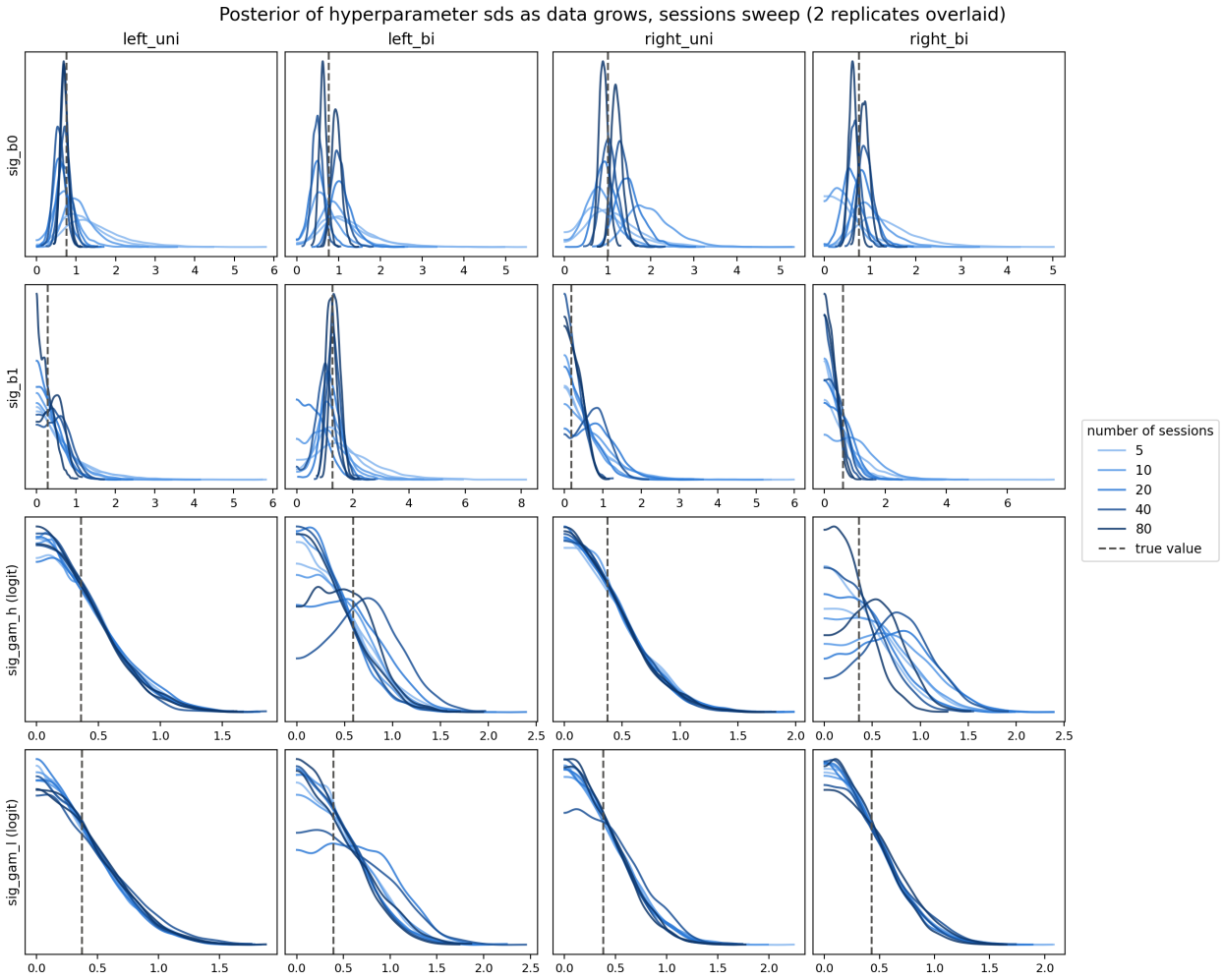


Figure 8: Posteriors of sd hyperparameters as sessions vary

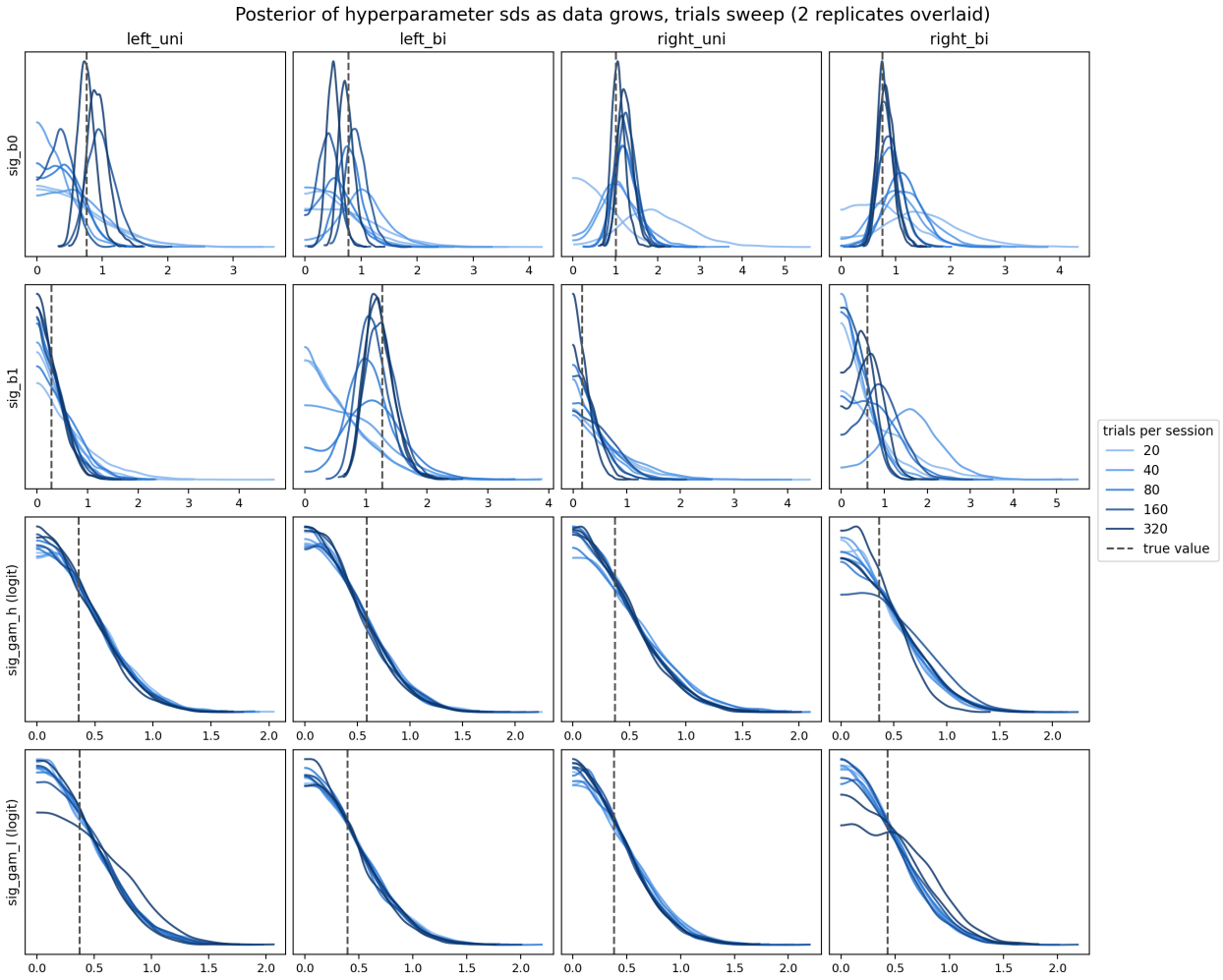


Figure 9: Posteriors of sd hyperparameters as trials vary

3.6 Earlier versions of this study

An earlier version of this study used hand-picked true values instead of the real-data fit: group slopes of 3 (left hand) and 6 (right hand), and between-session sds of the slope and lapse rates near 0. It was run under both the earlier and the current priors, on the same synthetic datasets. That comparison motivated the current priors: they removed the sampling problems of the smallest (5-session) datasets and the truncation of steep slopes at 8 imposed by the earlier $\mathcal{U}[0, 8]$ prior on μ_{β_1}. Its results are kept in `Results_Convergence_B_handpicked.pkl` (current priors) and `Results_Convergence_B_oldpriors.pkl` (earlier priors). The study reported here replaces it because the hand-picked values did not resemble the real data, in particular in the slopes and in the between-session sd of the slope.

3.7 Conclusions for the real data

At the size of the real dataset (42 sessions of about 308 trials), and at true values like those estimated from it:

- **Reliable:** per-session PSEs (RMSE about 1.5 μm, coverage 0.92–0.98) and JNDs; the group

means μ_{β_0} and μ_{β_1}; the bias spread σ_{β_0}; the slope spread σ_{β_1} when it is as large as that of left_bi.

- **Weakly determined:** the lapse-rate means, especially μ_{γ_h} of the unimanual groups, whose γ_h is near 0, and smaller slope spreads σ_{β_1}, which cannot be told from 0.
- **Biased in some datasets:** μ_{γ_l} of right_uni (too high), and the lapse rates whose side of the curve does not reach its asymptote: γ_h of left_bi (too low) and γ_l of left_uni (too high).
- **Session lapse rates are overconfident:** because σ_{γ_h} is not identified, a miss in a group's γ_h carries over to every session, and the bimanual session γ_h HDIs cover the truth in only 57% (left) and 86% (right) of sessions. The distractor's effect on γ_h in the real data should be read at the group level, not session by session.
- **Not identified:** σ_{γ_h} and σ_{γ_l}. Their posteriors largely reflect the priors and should not be interpreted.
- The sampler had no difficulties: no divergences in 20 fits.

Possible follow-ups.

1. Find out why μ_{γ_l} for right_uni is overestimated, here and in the earlier study.
2. Consider removing the session-level variation of the lapse rates (one γ_h and γ_l per group), since σ_{γ_h} and σ_{γ_l} are not identified and the session lapse rates are overconfident.
3. For future experiments: stimulus levels further out would let the curves reach their asymptotes and identify the lapse rates.